

AI and Generative AI in Education: Critical Understanding and Application

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1. Understanding AI and Generative AI

Artificial Intelligence (AI) refers to computer systems designed to perform tasks that typically require human cognition, such as logical reasoning, pattern recognition, decision-making, and problem-solving. Generative AI (GenAI) is an advanced subset of AI capable of creating entirely new, original content—including text, images, computer code, and audio—by learning patterns from massive datasets rather than simply processing existing data.

In educational contexts, these technologies serve distinct purposes:

- **Adaptive Learning Platforms (Traditional AI):** Systems such as Century Tech use machine learning algorithms to evaluate student performance in real time. The system identifies knowledge gaps and automatically adjusts the difficulty of mathematical problems to match individual learning speeds.
- **Dynamic Content Generation (Generative AI):** Educators utilize tools like ChatGPT or Claude to generate customized differentiated reading passages, produce structured lesson plans, or create rubrics tailored to specific learning outcomes.
- **Automated Feedback & Chatbots:** AI-powered virtual assistants provide learners with instant support on routine queries, while automated assessment tools assist teachers by analyzing written essays for grammatical structure and thematic clarity.

2. AI and Education

Key Premise: The integration of Artificial Intelligence (AI) in education provides significant opportunities for enhanced learning alongside complex ethical challenges.

On the positive side, AI enables the hyper-personalisation of instruction, allowing students to progress at an optimal pace tailored to their unique learning styles. Furthermore, by automating administrative duties such as attendance tracking and preliminary grading, AI frees valuable time for educators to focus on direct student mentorship and emotional support.

Conversely, significant risks exist. Generative AI systems can produce factual inaccuracies ("hallucinations") and perpetuate historical or demographic biases present in their training data.

Over-reliance on automated tools also threatens to diminish critical thinking skills among learners and reduce authentic human interaction within classrooms.

Research by UNESCO (2023) emphasizes that while AI offers powerful tools for educational equity, robust regulatory frameworks and teacher training are essential to safeguard human agency and data privacy. Ultimately, AI must complement, rather than replace, pedagogical expertise and human judgement.

Reference

UNESCO. (2023). *Guidance for generative AI in education and research*. Paris: UNESCO.

3. Reflective Evaluation of AI Output

Educational Purpose & Tool Selection

I utilized Google Gemini to design a 45-minute inquiry-based lesson plan and a accompanying five-question diagnostic quiz focused on introductory cellular biology for secondary school students.

Observation of Bias, Hallucinations, and Errors

While the structural outline of the lesson plan was logical, critical evaluation revealed subtle inaccuracies. First, the quiz contained a hallucinated assertion claiming that animal cells contain cell walls, which is factually incorrect. Second, the generated discussion prompts displayed contextual bias by assuming all students had access to digital laboratory equipment, overlooking schools with limited resource availability.

Verification of Accuracy and Fairness

To rectify these issues, I cross-checked the scientific definitions and quiz items against standardized biology textbooks and peer-reviewed educational repositories. I modified the biased activities by incorporating low-cost, hands-on alternatives suitable for diverse classroom settings.

Importance of Critical Evaluation

Critically evaluating AI output is paramount in educational settings. AI models generate text based on probabilistic patterns rather than true comprehension. Accepting outputs without verification risks disseminating inaccurate information and reinforcing implicit social biases. Educators must retain full ownership of the pedagogical process, ensuring that technological tools strictly align with ethical standards and diverse learning needs.

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